

- 1 Three liquid fractions (P, Q, and R) are extracted from a biological sample. Their physical properties are shown in the table below.

| Fraction | Density / $\text{g cm}^{-3}$ | Boiling point / $^{\circ}\text{C}$ | Miscibility             |
|----------|------------------------------|------------------------------------|-------------------------|
| P        | 0.79                         | 56                                 | Miscible with Q         |
| Q        | 1.00                         | 100                                | Miscible with P         |
| R        | 1.49                         | 61                                 | Immiscible with P and Q |

The fractions are mixed in a separating funnel and allowed to settle, forming two layers.

Which procedure describes the correct method to obtain a pure sample of fraction P?

- A Collect the upper layer and heat to  $56^{\circ}\text{C}$   
B Collect the upper layer and heat to  $100^{\circ}\text{C}$   
C Collect the lower layer and heat to  $56^{\circ}\text{C}$   
D Collect the lower layer and heat to  $61^{\circ}\text{C}$   
E Collect the upper layer and heat to  $61^{\circ}\text{C}$
- 2 A student conducts two experiments to heat liquids using the energy released from the combustion of a fixed amount of ethanol. Assume that there is no heat loss and that the amount of energy transferred to the liquid is the same in both experiments.

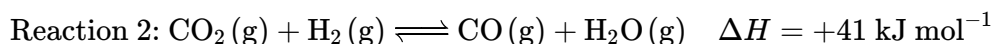
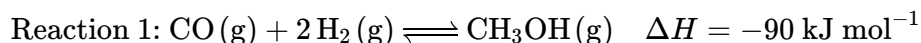
In Experiment 1, mass  $m$  of liquid X (specific heat capacity  $c$ ) is heated, and the temperature rises by  $\Delta T$ .

In Experiment 2, mass  $3m$  of liquid Y (specific heat capacity  $\frac{1}{2}c$ ) is heated.

Which expression gives the temperature rise in Experiment 2?

- A  $\frac{1}{6}\Delta T$   
B  $\frac{1}{3}\Delta T$   
C  $\frac{2}{3}\Delta T$   
D  $\frac{3}{2}\Delta T$   
E  $6\Delta T$

- 3 Two reversible gas-phase reactions are carried out in separate closed vessels:



The following actions could be applied independently to each reaction system:

- 1 decreasing the volume of the reaction container at constant temperature
- 2 increasing the temperature of the reaction container at constant volume
- 3 adding an appropriate catalyst at constant temperature and volume

Assuming all gases behave ideally, which of these actions will increase the initial rate of the forward reaction for **both** reactions, **and** increase the equilibrium yield of products for Reaction 1, while having **no effect** on the equilibrium yield of products for Reaction 2?

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 4 Which of the following statements about greenhouse gases is/are correct?

1. Methane can be released by the decomposition of organic waste in landfill sites.
2. Greenhouse gases mainly cool the Earth by reflecting incoming visible light back into space.
3. Burning fossil fuels can increase atmospheric carbon dioxide and strengthen the greenhouse effect.

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

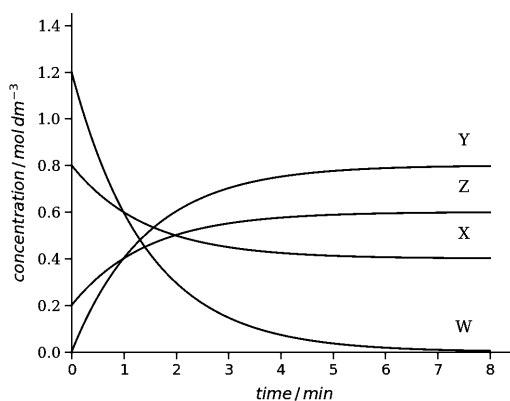
- 5 The complete combustion of the amino acid cysteine,  $\text{C}_3\text{H}_7\text{NO}_2\text{S}$ , produces carbon dioxide, water, nitrogen gas, and sulfur dioxide according to the equation:



What is the value of  $b$  when the equation is balanced using the lowest whole-number coefficients?

- A 9  
B 11  
C 15  
D 17  
E 19  
F 23  
G 38
- 6 The graph shows how the concentrations of four substances, W, X, Y, and Z, change over time during a reaction in a closed vessel at constant temperature.

Which of the following is the balanced equation for this reaction?

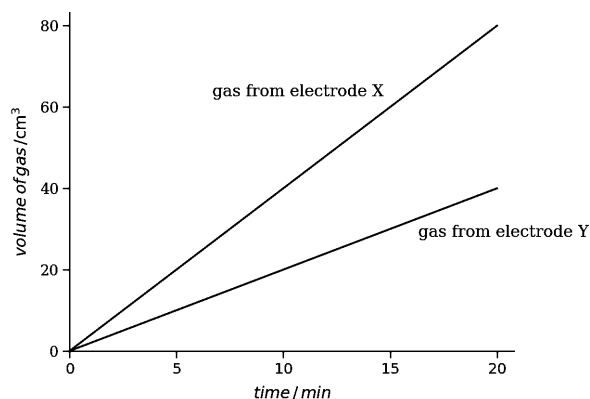


- A  $3\text{W} + \text{X} \rightarrow 2\text{Y} + \text{Z}$   
B  $3\text{W} + \text{X} \rightarrow 4\text{Y} + 3\text{Z}$   
C  $3\text{W} + 2\text{X} \rightarrow 2\text{Y} + \text{Z}$   
D  $3\text{W} + 2\text{X} \rightarrow 2\text{Y} + 3\text{Z}$   
E  $6\text{W} + 4\text{X} \rightarrow 4\text{Y} + 3\text{Z}$   
F  $3\text{W} \rightarrow \text{X} + 2\text{Y} + \text{Z}$

- 7 The graph shows the volume of gas collected at each of two inert electrodes, X and Y, during the electrolysis of dilute aqueous sulfuric acid,  $\text{H}_2\text{SO}_4(\text{aq})$ , at constant current, temperature, and pressure.

Which of the following statements is/are correct?

1. Electrode X is connected to the negative terminal of the power supply.
2. In the external circuit, electrons flow from electrode X to electrode Y.
3. For every 1 mol of gas collected at electrode Y, 4 mol of electrons pass through the external circuit.



- A** none of them  
**B** 1 only  
**C** 2 only  
**D** 3 only  
**E** 1 and 2 only  
**F** 1 and 3 only  
**G** 2 and 3 only  
**H** 1, 2 and 3

- 8 The following information about four non-metallic elements labelled W, X, Y and Z is given:
- Elements W and Y displace bromine from an aqueous solution of potassium bromide,  $\text{KBr(aq)}$ .
  - Elements X and Z cannot displace bromine from an aqueous solution of potassium bromide,  $\text{KBr(aq)}$ .
  - Element Y reacts explosively with hydrogen gas in the dark, whereas element W requires ultraviolet light to react with hydrogen gas.
  - Element Z reacts with an aqueous solution of the sodium salt  $\text{NaX}$  to displace element X.

What is the order of reactivity (oxidising ability) of these four elements, starting with the most reactive?

- A W, Y, X, Z
  - B W, Y, Z, X
  - C X, Z, W, Y
  - D X, Z, Y, W
  - E Y, W, X, Z
  - F Y, W, Z, X
  - G Z, X, W, Y
  - H Z, X, Y, W
- 9 Element X has an atomic number of 15. It reacts with hydrogen to form a simple molecular hydride.
- Which option gives the correct formula of the hydride and the shape of the molecule?
- A  $\text{XH}_2$ , linear
  - B  $\text{XH}_2$ , bent (V-shaped)
  - C  $\text{XH}_3$ , trigonal planar
  - D  $\text{XH}_3$ , trigonal pyramidal
  - E  $\text{XH}_4$ , tetrahedral

**10** Calcium carbonate reacts with hydrochloric acid according to the equation:



In a baseline experiment (represented by curve **X** on the graph), 2.00 g of large marble chips ( $\text{CaCO}_3$ ) reacts with  $50 \text{ cm}^3$  of  $1.0 \text{ mol dm}^{-3}$   $\text{HCl}(\text{aq})$  at  $25^\circ\text{C}$ .

Two further experiments are carried out at  $25^\circ\text{C}$ :

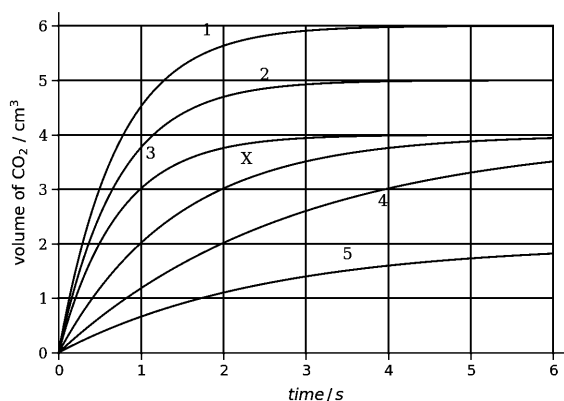
- **Experiment 1:** 3.00 g of powdered  $\text{CaCO}_3$  is reacted with  $50 \text{ cm}^3$  of  $1.0 \text{ mol dm}^{-3}$   $\text{HCl}(\text{aq})$

- **Experiment 2:** 2.00 g of large marble chips ( $\text{CaCO}_3$ ) is reacted with  $100 \text{ cm}^3$  of  $0.40 \text{ mol dm}^{-3}$   $\text{HCl}(\text{aq})$

( $M_r$  value:  $\text{CaCO}_3 = 100$ )

Which curves show how the volume of  $\text{CO}_2$  evolved changes with time in Experiment 1 and Experiment 2?

|          | Experiment 1 | Experiment 2 |
|----------|--------------|--------------|
| <b>A</b> | curve 1      | curve 4      |
| <b>B</b> | curve 1      | curve 5      |
| <b>C</b> | curve 2      | curve 4      |
| <b>D</b> | curve 2      | curve 5      |
| <b>E</b> | curve 3      | curve 4      |
| <b>F</b> | curve 3      | curve 5      |

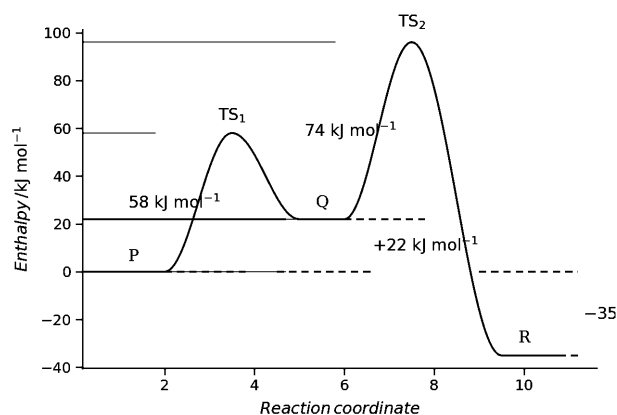


- 11 The reaction energy profile for a two-step gas-phase reaction  $P(g) \rightarrow Q(g) \rightarrow R(g)$  is shown in the diagram below.

From the information provided in the energy profile:

- The activation energy for the forward step  $P \rightarrow Q$  is  $58 \text{ kJ mol}^{-1}$ .
- The intermediate  $Q$  has an enthalpy  $22 \text{ kJ mol}^{-1}$  higher than reactant  $P$ .
- The activation energy for the forward step  $Q \rightarrow R$  is  $74 \text{ kJ mol}^{-1}$ .
- The overall enthalpy change for  $P \rightarrow R$  is  $\Delta H = -35 \text{ kJ mol}^{-1}$ .

What are the activation energy for the conversion of intermediate  $Q$  to reactant  $P$  ( $E_a(Q \rightarrow P)$ ) and the activation energy for the conversion of product  $R$  to intermediate  $Q$  ( $E_a(R \rightarrow Q)$ )?



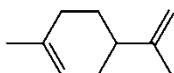
- A  $E_a(Q \rightarrow P) = 36 \text{ kJ mol}^{-1}$ ,  $E_a(R \rightarrow Q) = 61 \text{ kJ mol}^{-1}$
- B  $E_a(Q \rightarrow P) = 36 \text{ kJ mol}^{-1}$ ,  $E_a(R \rightarrow Q) = 109 \text{ kJ mol}^{-1}$
- C  $E_a(Q \rightarrow P) = 36 \text{ kJ mol}^{-1}$ ,  $E_a(R \rightarrow Q) = 131 \text{ kJ mol}^{-1}$
- D  $E_a(Q \rightarrow P) = 58 \text{ kJ mol}^{-1}$ ,  $E_a(R \rightarrow Q) = 109 \text{ kJ mol}^{-1}$
- E  $E_a(Q \rightarrow P) = 58 \text{ kJ mol}^{-1}$ ,  $E_a(R \rightarrow Q) = 131 \text{ kJ mol}^{-1}$
- F  $E_a(Q \rightarrow P) = 80 \text{ kJ mol}^{-1}$ ,  $E_a(R \rightarrow Q) = 39 \text{ kJ mol}^{-1}$
- G  $E_a(Q \rightarrow P) = 80 \text{ kJ mol}^{-1}$ ,  $E_a(R \rightarrow Q) = 109 \text{ kJ mol}^{-1}$
- H  $E_a(Q \rightarrow P) = 80 \text{ kJ mol}^{-1}$ ,  $E_a(R \rightarrow Q) = 131 \text{ kJ mol}^{-1}$
- 12 A pure liquid is boiling at its normal boiling point while energy is supplied continuously. Why can the temperature remain constant while boiling continues?
- A The particles stop moving while bonds are broken.
- B The supplied energy is destroyed as the gas forms.
- C The supplied energy is used to overcome attractions between particles as they separate.
- D The particles lose kinetic energy as the liquid boils.
- E The mass of each particle decreases during boiling.

**13** The skeletal structure of limonene is shown below:

Limonene undergoes complete catalytic hydrogenation to form a fully saturated hydrocarbon.

What is the minimum volume of hydrogen gas, measured at room temperature and pressure (RTP), required to react completely with 1.60 cm<sup>3</sup> of limonene?

( $M_r$  of limonene = 136; density of limonene = 0.85 g cm<sup>-3</sup>; molar volume of a gas at RTP = 24.0 dm<sup>3</sup> mol<sup>-1</sup>)



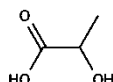
- A** 0.024 dm<sup>3</sup>
- B** 0.048 dm<sup>3</sup>
- C** 0.24 dm<sup>3</sup>
- D** 0.48 dm<sup>3</sup>
- E** 0.72 dm<sup>3</sup>
- F** 0.96 dm<sup>3</sup>
- G** 2.4 dm<sup>3</sup>
- H** 4.8 dm<sup>3</sup>

- 14** A 0.45 g sample of the organic compound shown below reacts completely with an excess of sodium metal at room temperature.

Any gas produced is collected and its volume is measured at room temperature and pressure (r.t.p.).

Assuming 1.0 mol of gas occupies  $24.0 \text{ dm}^3$  at r.t.p., what volume of gas is collected?

( $A_r$  values: H = 1.0; C = 12.0; O = 16.0)



- A**  $0.12 \text{ cm}^3$   
**B**  $60 \text{ cm}^3$   
**C**  $120 \text{ cm}^3$   
**D**  $180 \text{ cm}^3$   
**E**  $240 \text{ cm}^3$   
**F**  $360 \text{ cm}^3$   
**G**  $600 \text{ cm}^3$
- 15** An aqueous solution gives a green precipitate when aqueous sodium hydroxide is added. A separate sample gives a white precipitate when aqueous barium chloride is added in the presence of dilute hydrochloric acid. Which salt could the solution contain?
- A**  $\text{FeSO}_4$   
**B**  $\text{FeCl}_2$   
**C**  $\text{CuSO}_4$   
**D**  $\text{Fe}_2(\text{SO}_4)_3$   
**E**  $\text{MgSO}_4$   
**F**  $\text{CaCl}_2$

**16** Which of the following are common properties of transition metals?

1. They can form stable ions in different oxidation states.
2. Their compounds are often coloured.
3. They or their compounds are often used as catalysts.

- A** none of them  
**B** 1 only  
**C** 2 only  
**D** 3 only  
**E** 1 and 2 only  
**F** 1 and 3 only  
**G** 2 and 3 only  
**H** 1, 2 and 3

**17** An experiment investigates the reaction between solid marble chips (calcium carbonate,  $\text{CaCO}_3$ ) and an excess of dilute hydrochloric acid:



In a second experiment, the same mass of  $\text{CaCO}_3$  is used, but the solid is ground into a fine powder. All other conditions, including the volume, concentration, and initial temperature of the hydrochloric acid, are kept exactly the same.

How do the frequency of collisions between reactant particles, the activation energy of the reaction, and the initial rate of reaction in the second experiment compare to those in the first experiment?

|          | Frequency of collisions between reactant particles | Activation energy | Initial rate of reaction |
|----------|----------------------------------------------------|-------------------|--------------------------|
| <b>A</b> | decreases                                          | decreases         | decreases                |
| <b>B</b> | decreases                                          | unchanged         | decreases                |
| <b>C</b> | unchanged                                          | decreases         | increases                |
| <b>D</b> | unchanged                                          | unchanged         | increases                |
| <b>E</b> | increases                                          | decreases         | increases                |
| <b>F</b> | increases                                          | increases         | increases                |
| <b>G</b> | increases                                          | unchanged         | increases                |
| <b>H</b> | increases                                          | unchanged         | unchanged                |

- 18** Concentrated aqueous solutions of the following three compounds are electrolysed separately using inert electrodes: 1. NaCl 2. HBr 3. CuCl<sub>2</sub>

Which of the solutions yield(s) the elemental form of **all** the constituent elements of the solute at the electrodes?

- A** 3 only
- B** 1 and 2 only
- C** 1 and 3 only
- D** 2 and 3 only
- E** 1, 2 and 3

- 19** A redox reaction occurs in acidic solution between the complex ion [PtCl<sub>4</sub>]<sup>2-</sup> and the permanganate ion MnO<sub>4</sub><sup>-</sup>. In this reaction:

\* The platinum is oxidised from its state in the complex ion to the +4 oxidation state. \* The manganese is reduced from its state in the permanganate ion to the +2 oxidation state.

What is the simplest whole number mole ratio of [PtCl<sub>4</sub>]<sup>2-</sup> to MnO<sub>4</sub><sup>-</sup> required for the reaction?

- A** 1 : 1
- B** 2 : 1
- C** 2 : 5
- D** 5 : 2
- E** 5 : 1

- 20** A sample of a metal M consists of only two isotopes: <sup>69</sup>M (relative isotopic mass 69.0) and <sup>71</sup>M (relative isotopic mass 71.0).

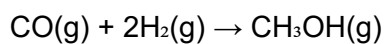
A 4.872 g sample of pure metal M reacts completely with oxygen to form 6.552 g of the oxide M<sub>2</sub>O<sub>3</sub>.

What is the percentage abundance of the <sup>69</sup>M isotope in this sample?

(Assume the relative atomic mass of oxygen is 16.0.)

- A** 30%
- B** 35%
- C** 60%
- D** 65%
- E** 70%
- F** 75%

- 21** Methanol can be synthesised industrially in the gas phase from carbon monoxide and hydrogen according to the following equation:



When 1 mol of CO(g) reacts completely with hydrogen, 112 kJ of energy is released.

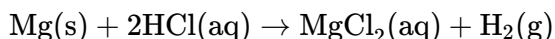
Some mean bond enthalpies are given in the table below:

| Bond | Mean bond enthalpy / $\text{kJ mol}^{-1}$ |
|------|-------------------------------------------|
| C-H  | 413                                       |
| C-O  | 358                                       |
| O-H  | 464                                       |
| H-H  | 436                                       |

What is the bond enthalpy of the carbon–oxygen bond in the carbon monoxide molecule?

- A** 613  $\text{kJ mol}^{-1}$
- B** 965  $\text{kJ mol}^{-1}$
- C** 1077  $\text{kJ mol}^{-1}$
- D** 1189  $\text{kJ mol}^{-1}$
- E** 1301  $\text{kJ mol}^{-1}$
- F** 1513  $\text{kJ mol}^{-1}$

- 22** Magnesium reacts with hydrochloric acid according to the equation:



Five experiments are set up to produce hydrogen gas.

| Exp | Mass Mg / g | Vol HCl / cm <sub>3</sub> | Conc HCl / mol dm <sup>-3</sup> | State of Mg | Temp / °C |
|-----|-------------|---------------------------|---------------------------------|-------------|-----------|
| 1   | 1.2         | 200                       | 2.0                             | Powder      | 60        |
| 2   | 3.6         | 150                       | 1.0                             | Powder      | 50        |
| 3   | 2.4         | 300                       | 1.0                             | Ribbon      | 20        |
| 4   | 2.4         | 100                       | 2.0                             | Powder      | 40        |
| 5   | 4.8         | 100                       | 2.0                             | Ribbon      | 40        |

(Ar value: Mg = 24)

Which experiment produces at least 0.10 mol of H<sub>2</sub> in the shortest time?

- A** Experiment 1
  - B** Experiment 2
  - C** Experiment 3
  - D** Experiment 4
  - E** Experiment 5
- 23** An organic carboxylic acid **X** has the condensed structural formula HOOCCH<sub>2</sub>CH<sub>2</sub>COOH. Acid **X** contains two carboxylic acid groups.

A student dissolves 0.590 g of acid **X** in distilled water and makes the volume up to 250 cm<sup>3</sup> in a volumetric flask.

A 25.0 cm<sup>3</sup> portion of this solution is transferred to a conical flask and titrated against aqueous potassium hydroxide, KOH(aq). Exactly 20.0 cm<sup>3</sup> of the KOH(aq) solution is required for complete neutralisation.

What is the concentration of the potassium hydroxide solution in mol dm<sup>-3</sup>?

(A<sub>r</sub> values: H = 1.0, C = 12.0, O = 16.0)

- A** 0.0125 mol dm<sup>-3</sup>
- B** 0.0250 mol dm<sup>-3</sup>
- C** 0.0500 mol dm<sup>-3</sup>
- D** 0.100 mol dm<sup>-3</sup>
- E** 0.250 mol dm<sup>-3</sup>
- F** 0.500 mol dm<sup>-3</sup>

- 24** Propane-1,3-diol,  $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{OH}$ , reacts with butanedioic acid,  $\text{HOOCCH}_2\text{CH}_2\text{COOH}$ , to form a polyester.

Which expression represents a repeating unit of the polyester?

- A**  $[-\text{OCH}_2\text{CH}_2\text{CH}_2\text{OCOCH}_2\text{CH}_2\text{CO}-]_n$   
**B**  $[-\text{CH}_2\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_2\text{COO}-]_n$   
**C**  $[-\text{NHCH}_2\text{CH}_2\text{CH}_2\text{NHCOCH}_2\text{CH}_2\text{CO}-]_n$   
**D**  $[-\text{OCH}_2\text{CH}_2\text{CH}_2\text{OCH}_2\text{CH}_2\text{CH}_2\text{O}-]_n$   
**E**  $[-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2-]_n$

- 25** Element  $X$  is in Period 3 of the Periodic Table. In its ground state, an atom of  $X$  has three unpaired electrons, and it reacts with chlorine to form a chloride with the formula  $\text{XCl}_3$ .

Which row of the table correctly describes the elements adjacent to  $X$  in the Periodic Table?

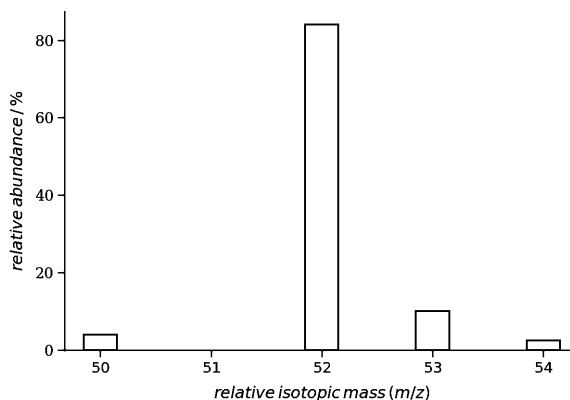
|   | element immediately above $X$                         | element immediately below $X$                    | element immediately to the left of $X$           | element immediately to the right of $X$       |
|---|-------------------------------------------------------|--------------------------------------------------|--------------------------------------------------|-----------------------------------------------|
| A | exists as diatomic molecules containing a triple bond | has a larger atomic radius than $X$              | forms an oxide with a giant covalent structure   | has a higher first ionisation energy than $X$ |
| B | can expand its octet to form a pentachloride          | has a larger atomic radius than $X$              | contains one more proton in its nucleus than $X$ | has a lower first ionisation energy than $X$  |
| C | exists as diatomic molecules containing a triple bond | has a larger atomic radius than $X$              | forms an oxide with a giant covalent structure   | has a lower first ionisation energy than $X$  |
| D | has a lower first ionisation energy than $X$          | contains fewer occupied electron shells than $X$ | forms an oxide with a giant covalent structure   | has a higher first ionisation energy than $X$ |
| E | exists as diatomic molecules containing a triple bond | has a higher first ionisation energy than $X$    | contains one more proton in its nucleus than $X$ | has a lower first ionisation energy than $X$  |

- 26** The solubility of solid X in water is shown. A saturated solution is prepared using 50 g of water at 60 °C and is then cooled to 20 °C. Assume no water evaporates.

What mass of X crystallises?

| temperature / $^{\circ}\text{C}$ | solubility of X / g per 100 g water |
|----------------------------------|-------------------------------------|
| 20                               | 30                                  |
| 60                               | 80                                  |

- A** 10 g  
**B** 15 g  
**C** 20 g  
**D** 25 g  
**E** 40 g
- 27** The mass spectrum of an element X is shown in the graph below.
- A 0.100 mol sample of element X reacts completely with 0.075 mol of  $\text{O}_2(\text{g})$  to form a single oxide. Naturally occurring oxygen consists of three isotopes with mass numbers 16, 17, and 18.
- Using this information, what is the difference between the maximum possible and the minimum possible relative formula mass of this oxide?



- A** 6  
**B** 8  
**C** 10  
**D** 12  
**E** 14  
**F** 16  
**G** 22